

Database Design & Data Model Specification

Pure MongoDB NoSQL Schema Architecture, Indexing Strategies & Document ODM Mappings

Production documentation artifact for the AI-Powered Household Appliance Rental System (RentAI / SDC2), encompassing enterprise Pure NoSQL architecture, customer churn prevention, multi-paradigm collaborative filtering, and local LLM context assistance.

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Project Title:	RentAI / SDC2 Household Appliance Ecosystem	Classification:	Confidential / Academic & Enterprise
Lead Architecture:	Deep Learning & Full-Stack Systems Team	Publication Date:	September 2026
Status:	VERIFIED & APPROVED	Target Platform:	Ubuntu 24.04 LTS / Windows 11 Enterprise

1. Architectural Rationale: Pure NoSQL (Zero SQL)

Traditional Relational Database Management Systems (RDBMS) enforce rigid columnar structures requiring costly ALTER TABLE schema migrations whenever new appliance categories are introduced (e.g. adding IoT power telemetry or water filter specifications). RentAI deliberately eliminated SQL databases (configured with `DATABASES = {}` in Django settings) in favor of **MongoDB 8.3** operated via **MongoEngine ODM**.

2. Document Collection Specifications

2.1 Collection: `appliances` (Catalog Inventory)

Field Name	BSON Type	Index	Description
<code>_id</code>	ObjectId	Primary	Unique system identifier
<code>name</code>	String	Text	Appliance commercial title
<code>category</code>	String	Single (Asc)	Appliance category (AC, Fridge, TV)
<code>brand</code>	String	Single (Asc)	Manufacturer brand name
<code>price_per_day</code>	Double	Single (Asc)	Daily subscription charge
<code>monthly_rent</code>	Double	None	Pre-calculated monthly rent
<code>deposit</code>	Double	None	Refundable escrow deposit
<code>available</code>	Boolean	Compound	Current rental availability flag
<code>location</code>	String	Compound	Warehouse city (Hyderabad, Pune)
<code>specs</code>	Dict	None	Dynamic metadata attributes

2.2 Compound Indexing Strategy

To guarantee sub-15ms search latencies across 1,000+ units, MongoDB compound indexes were established:

```
db.appliances.createIndex({ category: 1, available: 1, price_per_day: 1 });
db.appliances.createIndex({ location: 1, available: 1 });
db.bookings.createIndex({ user_id: 1, status: 1 });
db.churn_scores.createIndex({ tenant_id: 1, calculated_at: -1 });
```

3. Data Provenance & Synthetic Lineage Tracking

To ensure academic reproducibility, the platform integrates a dedicated `data_provenance` collection tracking the empirical vs. simulated origin of every database record, noting synthetic noise multipliers, source datasets (e.g. IKEA, Kaggle Churn), and seed scripts.